

The Mean Boundary-Layer Profile

Computation of the steady laminar compressible base flow in pyMack

pyMack technical documentation

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This document specifies how `pyMack` computes the steady, laminar, compressible flat-plate boundary layer whose profiles $\bar{U}$, $\bar{T}$, $\bar{\rho}$, $\bar{\mu}$ and their wall-normal derivatives are the coefficients of the linear stability problem. Every equation, boundary condition, coefficient, and default value is stated exactly as implemented.

Notation. Overbars denote mean (base-flow) quantities. A prime on a mean quantity denotes $D \equiv d/dy$, or $d/d\eta$ in the similarity coordinate, as stated in context. The table below collects the base-flow, transport, and scale symbols used here.

Base flow & transport	
$\bar{U}, \bar{T}, \bar{\rho}=1/\bar{T}$	mean velocity, temperature, density
$\bar{\mu}, \bar{\kappa}$	viscosity, conductivity
$C=\bar{\mu}/\bar{T}$, K , B	transport groups, Eq. (11)

Scales & parameters	
Me , Re , Pr , γ	Mach, Reynolds, Prandtl, c_p/c_v
R_g , a_e	specific gas constant, edge sound speed ($a_e^2=\gamma R_g T_e$)
$L^*=\sqrt{\nu_e x/U_e}$	Mack length, Eq. (7)
$R=\sqrt{Re_x}$, F	Mack Reynolds number, reduced frequency, Eq. (8)
α , $\alpha_L=\alpha L^*$, δ^* , θ	streamwise wavenumber, scaled wavenumber, displacement/momentum thickness
η	similarity (generalised) wall-normal coordinate, Eq. (14)

1 Governing equations and non-dimensionalisation

The flow is governed by the compressible Navier–Stokes equations, expressing conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas [1]:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (\text{mass}) \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \quad (\text{momentum}) \quad (2)$$

$$\rho c_p \frac{DT}{Dt} = \frac{Dp}{Dt} + \nabla \cdot (\kappa \nabla T) + \Phi, \quad (\text{energy}) \quad (3)$$

$$p = \rho R_g T, \quad (\text{state}) \quad (4)$$

where $\mathbf{u} = (u, v)$ is the velocity, ρ the density, T the temperature, p the pressure, μ the viscosity, κ the conductivity, c_p the specific heat, R_g the specific gas constant (plain R is reserved for the Mack Reynolds number, Eq. (8)), and $D/Dt = \partial_t + \mathbf{u} \cdot \nabla$ the material derivative. The remaining terms are the Newtonian viscous stress tensor and the dissipation function,

$$\boldsymbol{\tau} = \mu \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right), \quad \Phi = \boldsymbol{\tau} : \nabla \mathbf{u}. \quad (5)$$

Their closed forms are not required in what follows; it suffices that both are linear in the velocity gradients. Equations (1)–(4) are applied twice: once to obtain the steady mean flow below, and once in linearised form about that mean flow to obtain the disturbance equations of linear stability theory.

Non-dimensionalisation and default gas constants. Velocities are scaled by the edge speed U_e , temperature by T_e , density by ρ_e , viscosity by μ_e , and pressure by $\rho_e U_e^2$. Because the gas is perfect and the pressure is nearly uniform across the layer, $\bar{p} = 1/\bar{T}$ in these units. Three dimensionless groups govern the problem,

$$M_e = \frac{U_e}{a_e}, \quad a_e = \sqrt{\gamma R_g T_e}, \quad Re = \frac{\rho_e U_e L_{\text{ref}}}{\mu_e}, \quad Pr = \frac{\mu_e c_p}{\kappa_e}, \quad (6)$$

where a_e is the edge speed of sound. The default gas constants are the standard air values $\gamma = 1.4$, $Pr = 0.72$, Sutherland constant $S = 110.4$ K [2], and power-law exponent $\omega_\mu \approx 0.7$ when the power-law viscosity model is selected.

The Mack length and Reynolds scales. A flat-plate boundary layer grows as $\sqrt{x}$, so the natural local length scale at station x , following Mack [3], is

$$L^* = \sqrt{\frac{\nu_e x}{U_e}}, \quad \nu_e = \frac{\mu_e}{\rho_e}. \quad (7)$$

With $L_{\text{ref}} = L^*$ the Reynolds number becomes the square root of the conventional one, and a fixed physical frequency f maps to a reduced frequency F ; both enter the neutral-curve and N -factor computations:

$$R = \frac{U_e L^*}{\nu_e} = \sqrt{Re_x}, \quad Re_x = \frac{U_e x}{\nu_e}; \quad F = \frac{2\pi f \nu_e}{U_e^2}; \quad \alpha_L = \alpha L^*, \quad (8)$$

where α is the streamwise wavenumber. R is the abscissa of every neutral curve and α_L the ordinate. The solver itself is dimensionless; the conversions above are pure postprocessing.

2 The self-similar base flow

The mean flow is obtained from a boundary-value problem, not an eigenvalue problem: the data are prescribed at both ends of the domain ($y = 0$ and $y \rightarrow \infty$) and the solution is unique. Its unknowns are the steady laminar profiles $\bar{U}$, $\bar{T}$, $\bar{\rho}$, $\bar{\mu}$ and their wall-normal derivatives, which subsequently enter the linearised disturbance equations as known coefficients.

For a flat plate the mean flow is *self-similar*: profiles at different streamwise stations collapse onto a single curve in the stretched coordinate η . Introducing the dimensionless stream function $f(\eta)$ with $f' = \bar{U} = u/U_e$ (in this section a prime denotes $d/d\eta$), the momentum and energy equations (2)–(3) reduce under the boundary-layer approximation to the classical pair of coupled ordinary differential equations [1, 4],

$$(C f'')' + f f'' = 0, \quad (9)$$

$$(B \bar{T}')' + f \bar{T}' + (\gamma - 1) M_e^2 C (f'')^2 = 0, \quad (10)$$

with the transport groups

$$C = \frac{\mu/\mu_e}{T/T_e}, \quad K = \frac{\kappa}{c_p \mu_e}, \quad B = \frac{K}{T} \left(= \frac{C}{Pr} \text{ at constant } Pr \right), \quad (11)$$

where C is the Chapman–Rubesin parameter [5]. The final term of (10) represents aerodynamic heating: it elevates the near-wall temperature at high Mach number and couples the energy equation to the velocity field. The conductivity model is $\kappa = \mu c_p / Pr$, so that $\bar{\kappa} = \bar{\mu} / Pr$ at constant Pr ; the transport derivatives $d\mu/dT$, $d\kappa/dT$ are evaluated analytically from the selected viscosity law.

Boundary conditions. Equations (9)–(10) are closed by

$$\underbrace{f(0) = 0, f'(0) = 0}_{\text{no slip}}, \quad \underbrace{f'(\infty) = 1, \bar{T}(\infty) = 1}_{\text{match the edge}}, \quad \begin{cases} \bar{T}'(0) = 0 & \text{(adiabatic wall)} \\ \bar{T}(0) = T_w/T_e & \text{(isothermal wall)} \end{cases} \quad (12)$$

For an adiabatic wall the recovery temperature is the standard result $T_{aw}/T_e = 1 + \frac{1}{2}(\gamma - 1)\sqrt{Pr} M_e^2$ [1]. For a cone the edge state itself is computed first, from the Taylor–Maccoll equation behind the conical shock; this is a separate upstream calculation and is not detailed further here.

Numerical solution. Equations (9)–(10) with conditions (12) are solved as a five-state first-order system $[f, \bar{U}, \bar{U}', \bar{T}, \bar{T}']$ by fourth-order collocation with residual control and Newton iteration [7, 8], integrated to $\eta_{\max} \approx 40$, well beyond the boundary-layer edge. On any requested grid the solver returns $\bar{U}, \bar{U}', \bar{U}'', \bar{T}, \bar{T}', \bar{T}'', \bar{\rho}, \bar{\mu}, d\mu/dT, d^2\mu/dT^2, \bar{\kappa}, d\kappa/dT, d^2\kappa/dT^2$.

Figure 1 shows the computed profiles for a representative hypersonic case: $M_e = 6$, edge temperature $T_e = 54$ K, and an isothermal wall at $T_w = 300$ K, i.e. $T_w/T_e = 5.56$. Since the adiabatic recovery temperature at these conditions is $T_{aw} \approx 7.1 T_e$, the wall is cooled. The velocity rises monotonically to its edge value, while viscous heating produces an interior temperature maximum, $\bar{T} \approx 5.7 T_e$, just above the cooled wall. The dotted line marks the outermost generalised inflection point, $(\bar{\rho} \bar{U}')' = 0$, the seat of the inviscid instability identified by Lees & Lin [6].

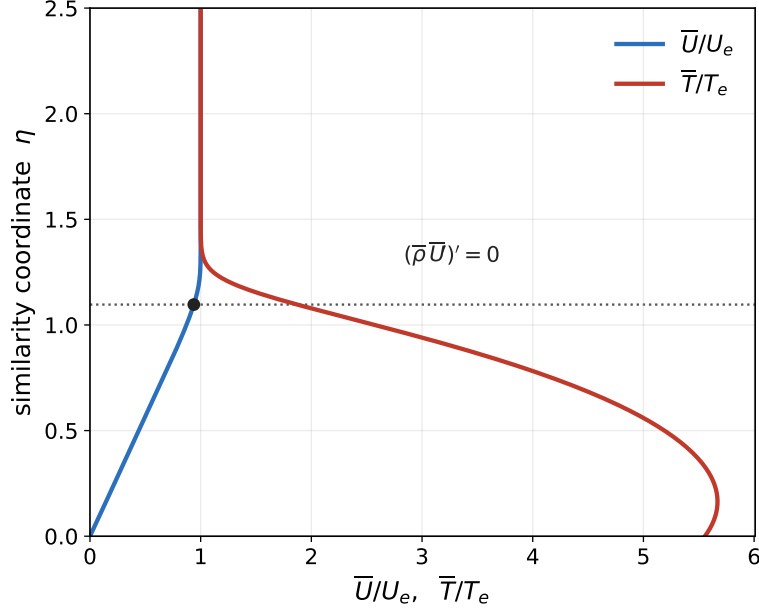


Figure 1: Computed compressible mean flow in the similarity coordinate η : $M_e = 6$, $T_e = 54$ K, isothermal wall at $T_w = 300$ K ($T_w/T_e = 5.56$, cooled relative to $T_{aw} \approx 7.1 T_e$). Viscous heating produces an interior temperature maximum $\approx 5.7 T_e$ just above the wall. The dotted line marks the outermost generalised inflection point $(\bar{\rho} \bar{U})' = 0$ [6]. These profiles and their wall-normal derivatives are the known coefficients of the linearised disturbance equations; the transformation to physical coordinates is given by Eqs. (14) and (15).

Remark: two equivalent formulations. Two equivalent self-similar formulations share the five-equation structure above. Equations (9)–(10) and the $\sqrt{2}$ transform of Section 3 constitute the classical compressible-Blasius form. The temporal solver uses instead the formulation of Özgen & Kırçalı [4], with that reference’s transport laws and normalisation ($f' = \bar{U}/\bar{T}$, factors of $\frac{1}{2}$, $y/L^* = \eta$, no $\sqrt{2}$). The two yield the same physical profiles; only the similarity bookkeeping differs.

3 Transformation to physical coordinates

The similarity solution is computed once, in η ; all case-specific dimensional information enters only through the transformation back to the physical wall-normal coordinate. Because the near-wall gas is hot ($\bar{T} > 1$) and correspondingly rarefied ($\bar{\rho} = 1/\bar{T}$), equal increments of η span unequal increments of physical distance. The map follows directly from the temperature field. In the units of Section 1 the differential relation is

$$\frac{dy}{d\eta} = \sqrt{2} \bar{T}(\eta) \quad (y \text{ in units of } L^*), \quad (13)$$

whose integral gives the physical coordinate at station x ,

$$\frac{y}{L^*} = \sqrt{2} \int_0^\eta \bar{T}(\eta') d\eta'. \quad (14)$$

Substituting the Mack length (7) yields the fully dimensional, case-specific form

$$y^*(\eta; x^*) = \sqrt{\frac{2 \nu_e x^*}{U_e}} \int_0^\eta \bar{T}(\eta') d\eta' \quad (15)$$

so that a single similarity solution generates the entire family of dimensional profiles, parameterised by the streamwise station x^* and the edge state (U_e, ν_e) . The map (14) is strictly monotone, and its inverse assigns a unique η to any physical height y^* . Since $\bar{T} > 1$ near the wall, the transform outpaces the linear reference $y/L^* = \sqrt{2}\eta$ there: a uniform η mesh maps to a physical mesh stretched into the hot near-wall region, which is why η , rather than y , is the natural computational coordinate. Wall-normal derivatives convert by the chain rule implied by (13),

$$\frac{d}{dy} = \frac{1}{\sqrt{2}\bar{T}(\eta)} \frac{d}{d\eta} \quad (y \text{ in units of } L^*). \quad (16)$$

The integral thicknesses follow from the same transform,

$$\frac{\delta^*}{L^*} = \sqrt{2} \int_0^\infty (\bar{T} - \bar{U}) d\eta, \quad \frac{\theta}{L^*} = \sqrt{2} \int_0^\infty \bar{U} (1 - \bar{U}) d\eta, \quad (17)$$

where the displacement thickness δ^* measures the outward displacement of the outer flow by the layer and the momentum thickness θ its momentum deficit. A profile may be stored on a y/δ^* or a y/L^* grid; the constant δ^*/L^* converts between them, with $D \mapsto (\delta^*/L^*)^{-1}D$.

The profiles $\bar{U}, \bar{T}, \bar{\rho}, \bar{\mu}$ and their wall-normal derivatives assembled here are the coefficients of the linearised disturbance equations of linear stability theory.

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